MIL-C-36416C
20 September 1978
SUPERSEDING
MIL-C-36416B
3 September 1976
MIL-C-0036416A(DSA-DM)
14 August 1972 and
MIL-C-36416(DM)
22 March 1965

MILITARY SPECIFICATION

CENTRIFUGE, LABORATORY, HEMATOLOGY, Small

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a small, compact, portable centrifuge for use by clinical laboratories in blood grouping, typing, cross matching, Coombs test and other cell washing procedures; it is suitable for operation on 110-120 volts, 60 Hz, AC.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-636	- Motors, Fractional Horsepower (Alternating Current).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-M-7298	- Manual, Technical: Commercial Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11	- Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

* UNDERWRITERS' LABORATORIES, INC.

UL 1262	- Laboratory Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746, 207 East Ohio Street, Chicago, IL 60611, and 1655 Scott Boulevard, Santa Clara, CA 95050).

3. REQUIREMENTS

3.1 Fire and casualty hazards.

* 3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 1262. Proof shall consist of one of the following:

a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

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b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts lists, all appropriately identified.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Material. Material shall be of the best quality to meet the functional requirements of the centrifuge. All hardware used shall be corrosion-resisting.

* 3.3 Design. The centrifuge shall consist of a base, and a rotating head with a surrounding guard and transparent plastic cover that are mounted above the base. The base shall be fabricated of cast iron and shall enclose the motor and electric components for driving the head. The base shall have cutouts for mounting the electric switch controls (see 3.4.3.1, 3.4.3.2 and 3.4.3.4). The bottom of the base shall be supplied with a removable metal base plate in order to allow access to the internal components. The base plate shall be provided with four cushioning rubber feet that are attached to the base by corrosion-resisting screws. The surfaces of the base plate shall be corrosion-resisting. A spindle shall be attached to the motor shaft and shall serve as a holder for the rotating head (see 3.4.1). The head shall provide easy, secure attachment to (and detachment from) the spindle. The surrounding guard shall be cylindrical in shape and shall be not less than 3/64 inch in thickness. The guard shall be constructed of aluminum alloy or corrosion-resisting steel. The plastic cover shall be not less than 1/8 inch in thickness, and shall be fabricated of methyl methacrylate or polycarbonate. The cover shall be hinged to the guard at one end, and a latch with manual release shall be provided to hold the closed cover in place. The latch shall prevent the cover from opening unless manually released. The cover shall have a hole (or holes) to allow for proper cooling of the spindle during rotation. The closed cover shall rest on a gasket that covers the exposed edge of the guard. The centrifuge shall be free of rocking and wobbling when placed on a flat surface. The overall weight of the centrifuge (including the head) shall not exceed 22 pounds. The overall height of the centrifuge shall not exceed 11-1/2 inches. The outside diameter of the guard shall be 8 inches $\pm$ 1/4 inch.

3.4 Components.

* 3.4.1 Head. A twelve-place rotating head, fabricated of lightweight impact-resistant plastic, shall be supplied. The tube slots shall be numbered from 1 to 12 and shall be capable of holding 10 x 75 millimeter (mm), 12 x 75 mm (Kahn)

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tubes, or other sizes not exceeding 12.7 x 75 mm. The head shall be readily removable from (or replaceable on) the spindle of the centrifuge; and shall also include a provision for indexing to permit return of the head to a matching board or holder after removal from the centrifuge. The head shall be designed to hold the tubes vertically or slightly tilted when it is stationary, and permit the bottoms of the tubes to swing out to an approximate angle of 45 degrees and a minimum radius of 2-3/4 inches during rotation of the head. The head shall also serve as a basket or transporting rack for holding and handling tubes. The head shall allow the use of an elastic band to hold the tubes in place, so that all tubes can be emptied simultaneously by inverting the head (e.g., pouring off supernatant liquid after centrifuging). The head shall be date-coded for easy age determination, and the service data (see 3.8) shall include a recommendation for automatic replacement of the head after five (5) years of use.

3.4.2 Motor. The motor shall be an induction type conforming to CC-M-636, class 3, and shall be suitable for operation on 115 ± 5 volts 60 Hz, AC supply lines. The motor shall have a minimum wattage rating of 15 watts, and shall be capable of driving a fully loaded head at two speeds that are specified in 3.4.3.2. The operation, speed selection and deceleration of the motor shall be controlled by the components specified in 3.4.3. The motor shall be provided with a blower to create an air stream through the housing for cooling purposes.

3.4.3 Motor controls.

3.4.3.1 Timer. The centrifuge motor shall be controlled by a three-minute rotary electric timer which includes a hold position for continuous operation. The timer shall have a base-mounted knob control with a calibrated dial that includes 5-second graduations from 0 to 1 minute and 15-second graduations from 1 to 3 minutes. When the motor is running, the time interval shall be capable of being shortened or lenghtened by adjusting the knob control accordingly. The position of the knob control shall indicate the amount of time left before the end of preselected time interval. The operation of the motor shall be shut off when the knob control is automatically returned to zero or manually set to zero.

3.4.3.2 Speed selector switch. The centrifuge motor shall be provided with a base-mounted toggle-type speed selector switch. The switch shall allow a choice of the following two motor speeds of operation:

a. A high speed of 3400 ± 150 revolutions per minute (rpm)
b. A low speed of 2400 ± 150 revolutions per minute (rpm).

The switch positions shall be identified by suitable markings to facilitate selection of the desired level of speed.

3.4.3.3 Centrifugal governor. The motor shall be provided with a centrifugal switch governor to control the motor speed when the speed selector switch of 3.4.3.2 is in the position for low speed operation.

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3.4.3.4 Brake. The centrifuge shall be provided with an electric brake circuit to shorten the deceleration time of the motor after the power to the motor is shut off. The brake circuit shall include a power resistor and a base-mounted toggle-type switch. The braking action shall be accomplished by holding the switch in the down position until the motor comes to rest. The position of the switch to accomplish braking shall be identified by a suitable marking.

3.4.3.5 Safety switch. A pressure-type safety switch shall be provided to control the electrical power to the timer so that the centrifuge will operate only when the cover is closed and securely latched.

3.4.4 Cord and plug assembly. A flexible power cord and non-detachable plug shall be provided with three conductors that terminate in a three-blade male plug that meets the requirements of ANSI C73-11. The exposed length of cord and plug shall not be less than 68 inches. The line cord shall be connected to the electric controls through a cutout in the base. Strain relief shall be provided at the cutout to secure the power cord and prevent abrasion to the cord insulation.

3.5 Performance.

3.5.1 Operational check.

3.5.1.1 Continuity of ground. The guard and base of the centrifuge shall provide grounding of metal surfaces for protection of personnel from shock. These shall be a positive indication of continuity of ground when tested as specified (see 4.4.1).

3.5.1.2 Motor operation. When tested as specified in 4.4.2:

a. The motor-driven head shall be capable of attaining the two modes of speed indicated in 3.4.3.2.

b. The centrifuge shall operate without any type of failure, or tendency to creep. There shall be no vibration or eccentricity that would indicate misalignment or imbalance of the motor or head.

c. The switch-controlled brake shall be capable of bringing the head to a stop in a maximum of 15 seconds.

d. The safety switch shall control the electric power to the timer.

3.5.1.3 Dielectric strength. There shall be no electrical breakdown when tested as specified (see 4.4.3).

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3.6 Finish. The exterior of the housing and the exterior and interior of the guard shall be painted. The painted surface shall be primed and baked. The primer and paint shall be applied in such a manner as to provide a uniform, chipproof, corrosion-resisting surface. The color shall be the manufacturer's standard for the item.

* 3.7 Identification marking. Each centrifuge shall bear a corrosion-resisting metal nameplate or foil conspicuously located, containing the following permanently marked data:

Name or registered trademark of the manufacturer
Serial number
Model number
Electrical characteristics

* 3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for a review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

* 3.9 Workmanship. Workmanship shall be first class throughout. Centrifuge and accessories shall be free from defects which detract from their appearance or impair their serviceability. All metal components shall be free of burrs, sharp edges, and other defects which may affect the safety of the operator. All internal electric wiring shall be adequately secure to preclude any breaking off at terminations. Any soldering that is used shall not result in cold solder joints. All mechanical terminations shall be adequate to preclude wires from breaking off or slipping out.

3.10 Accessories. Each centrifuge shall be supplied with twelve 10 x 75 mm (Kahn) tubes, for use with the head.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the motor, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with table II.

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TABLE II. For tests.

Characteristic	Requirement	Test procedure	Inspection level
Continuity of ground	3.5.1.1	4.4.1	100%
Motor operation	3.5.1.2	4.4.2	100%
Dielectric strength	3.5.1.3	4.4.3	100%

4.3 Examination. The centrifuge shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III classification of defects.

TABLE III. Classification of defects.

Category	Defects*
Major	
101	Centrifuge not complete with all component parts.
102	Components not usable or not capable of being properly secured to unit.
103	Centrifuge and accessories not free of burrs or sharp edges.
104	Rubber or gasket components not free of tears, cracks, or tackiness.
105	Internal wiring not adequately secured to terminations.
106	Centrifuge not free of corroded areas or rust spots.
107	Head component not properly securable to or removable from spindle.
108	Cover on guard not capable of being properly opened or closed.
109	Power cord not adequately secured within base.
110	Blades of plug for power cord not properly aligned.
111	Bottom plate or rubber feet not properly securable to the base.
112	Centrifuge not free of wobbling or rocking when placed on a flat surface.
Minor	
201	Identification marking not complete, legible or permanent.
202	Centrifuge not free of dirt, grease or removable foreign matter.
203	Finish not free of scratches or chipping.

* Examination shall not be restricted to the classified possible defects listed above.

* 4.3.2 Dimensional examination. The centrifuge shall be examined for defects in dimensions. Any dimension not within the limits specified herein shall be considered as a defect.

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4.3.3 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Continuity of ground. A continuity-indicating device (e.g., ohmmeter or battery and buzzer combination) shall be connected between the ground conductor in the plug and the centrifuge to assure that continuity exists between the ground conductor and the guard, base, and motor housing of the centrifuge.

* 4.4.2 Motor operation. The centrifuge head shall be loaded with twelve 75 x 10 mm or 75 x 12 mm (Kahn) tubes that are half filled with water. With the head in place, the centrifuge shall be operated at the high and low speed modes of operation (see 3.4.3.2) for a minimum of five minutes at each mode. At each speed selected, the motor shall reach maximum speed, and the speed of the head shall be determined by means of a stroboscope or a tachometer. In addition, at the high speed mode, the timer setting shall be set to zero to shut off the power to the motor, and the brake switch shall be manually depressed and held down until the head stops. The time for the head to stop rotating after the timer is set to zero shall be recorded.

* 4.4.3 Dielectric strength. The centrifuge shall be operated for a minimum of five minutes. Immediately after operation, with the power plug disconnected and the timer switch in the zero position, the centrifuge shall be subjected to a 60 Hz, AC potential gradually increasing to 1000 volts, that is applied between the current-carrying parts and metal enclosure, and maintained at maximum value for a period of one minute.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each centrifuge shall be thoroughly cleaned and dried in accordance with the applicable process as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

* 5.1.1.3 Unit package. Each centrifuge shall be clean, dry, wrapped in kraft paper or other suitable non-corrosive material. The centrifuge shall be further packaged in accordance with MIL-P-116, method III within the shipping container as specified in 5.2.1 or 5.2.2 as applicable.

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5.1.2 Level C. Centrifuge shall be clean dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

* 5.2.1 Level A. Each centrifuge shall be packed in a shipping container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class water resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

* 5.2.2 Level B. Each centrifuge shall be packed in a shipping container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged centrifuge shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers' rules and regulations.

* 5.3 Marking. Each unit package and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

* 6.1 Intended use. The Centrifuge, Laboratory, Hematology, is intended to be used by clinical laboratories in blood grouping, typing, cross matching, Coombs test and other cell washing procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.4 The Centrifuge covered by this specification is listed in the Federal Supply Catalog as follows:

National Stock Number	Item Identification
6640-00-145-1180	CENTRIFUGE, LABORATORY, SMALL, 115 Volt, 60 Hz, AC

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
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